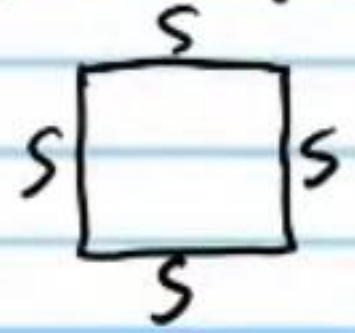
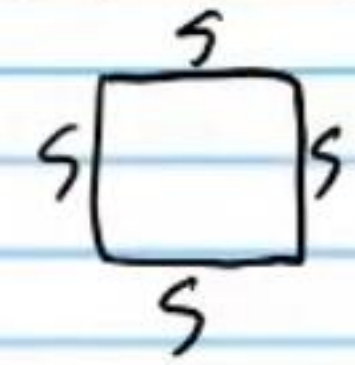


Class 35: Multistep Word Problems

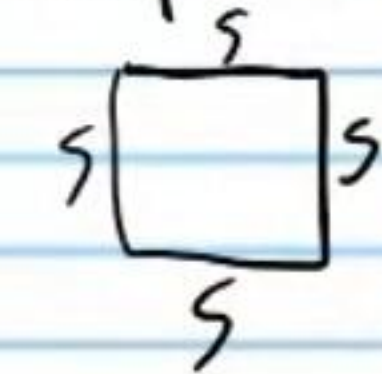
- 1A A square has perimeter 24. What is its area?



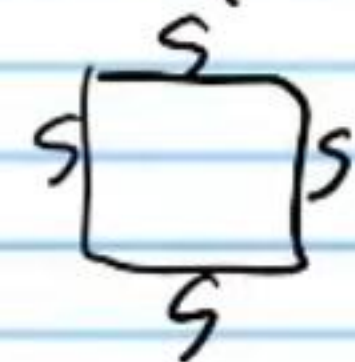
- 1B A square has an area of 64 square meters. What is its perimeter?



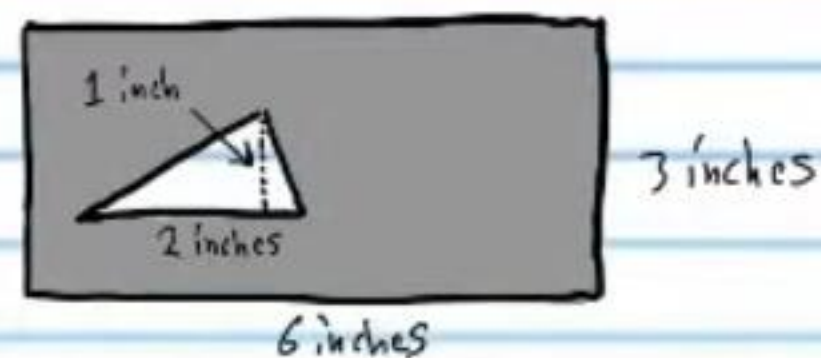
- 1C A square has perimeter 12. What is its area?



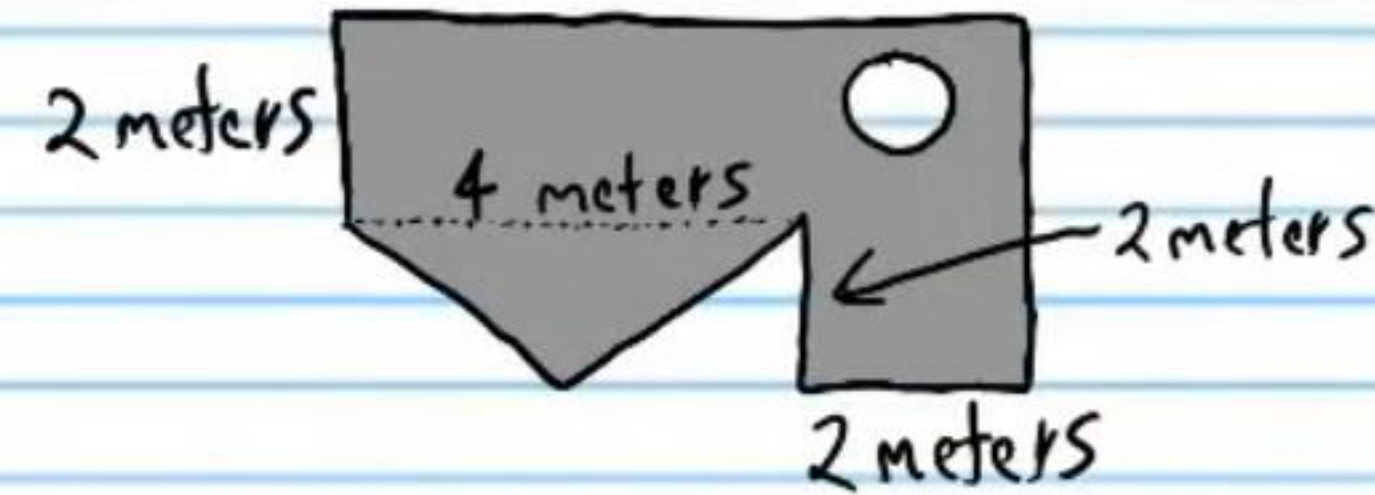
- 1D A square has area 81 in^2 . What is its perimeter?



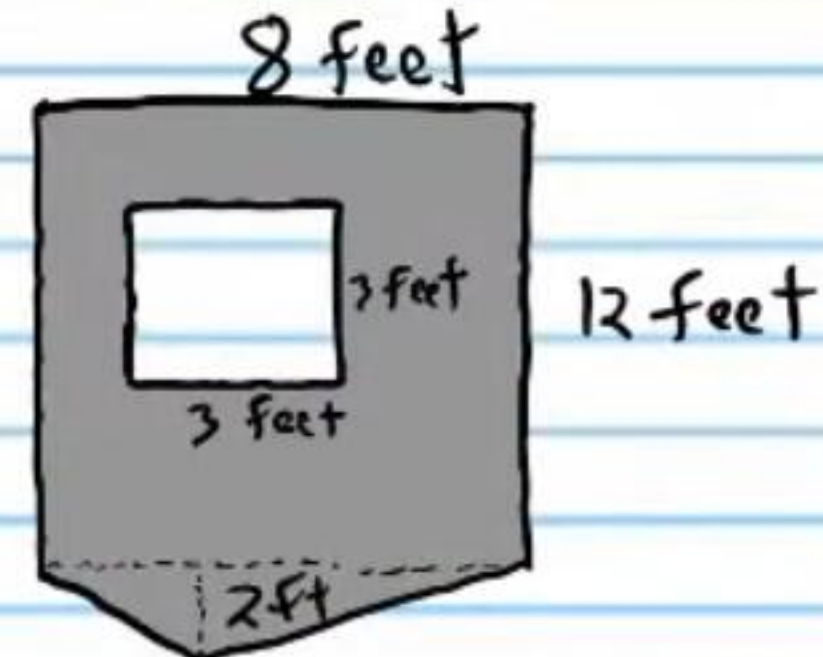
- 2A Find the area of the shaded region below.



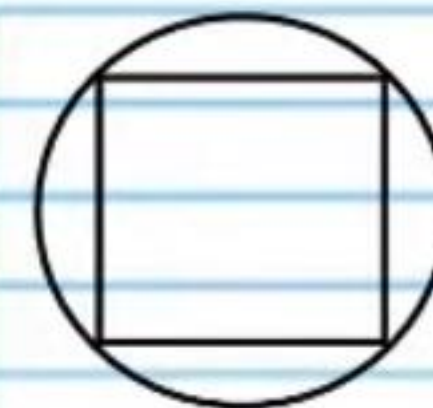
- 2B What is the area of the shaded region below if the area of the circle is 1 square meter?



- 2C Find the area of the shaded region below.



- 2D If the area of the circle below is 20 ft^2 , and the side length of the square is 4 ft, what is the area of one of the sections inside the circle but outside the square?



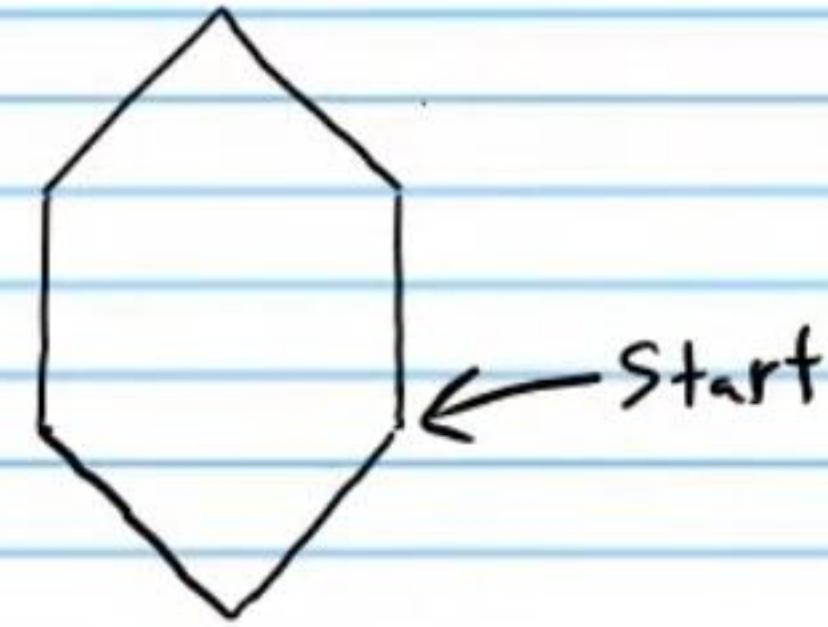
3A) If it takes three electricians to wire 36 microwaves in four hours, how many microwaves can five electricians wire in seven hours?

3B) It takes 7 days for 6 tailors to make 168 t-shirts. How many t-shirts can 5 tailors make in 3 days?

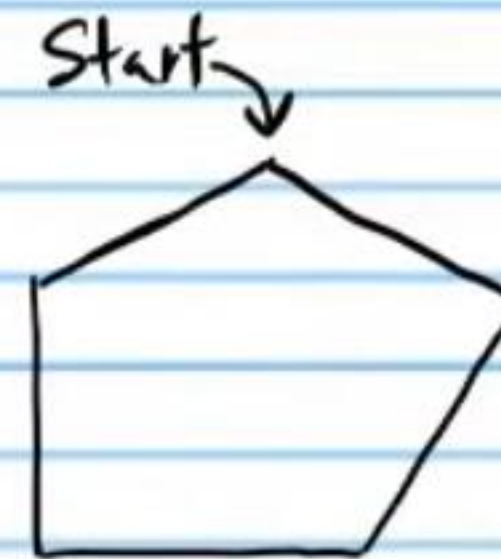
3C) If it takes 2 days for 4 assemblers to build 32 computers, how many computers can 5 assemblers build in 10 days?

3D) It takes two workers to build 10 machines in five days. How many machines can eight workers build in 3 days?

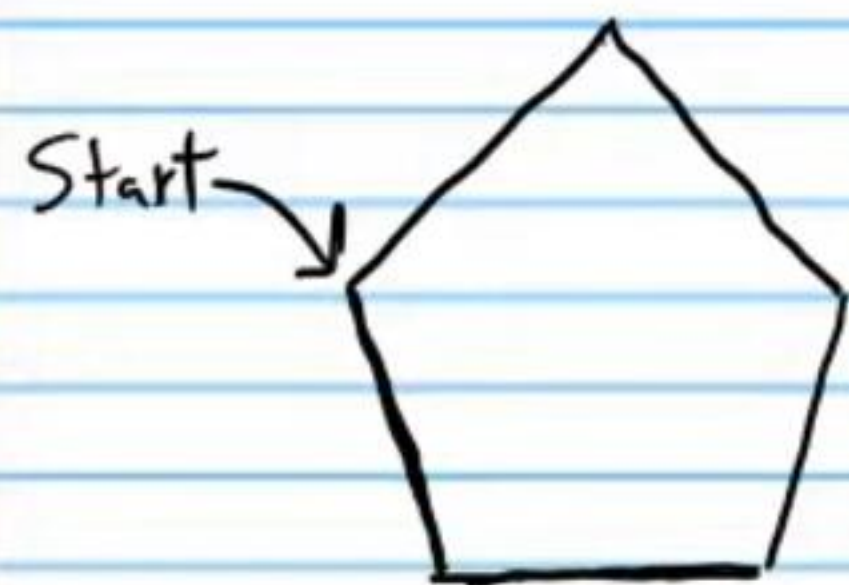
4A) An object travels on the sides of the shape below. It takes the object three seconds to move from one corner to the next. If the object starts on the bottom right corner, and moves clockwise, at which side will it be after traveling for 2,048 seconds?



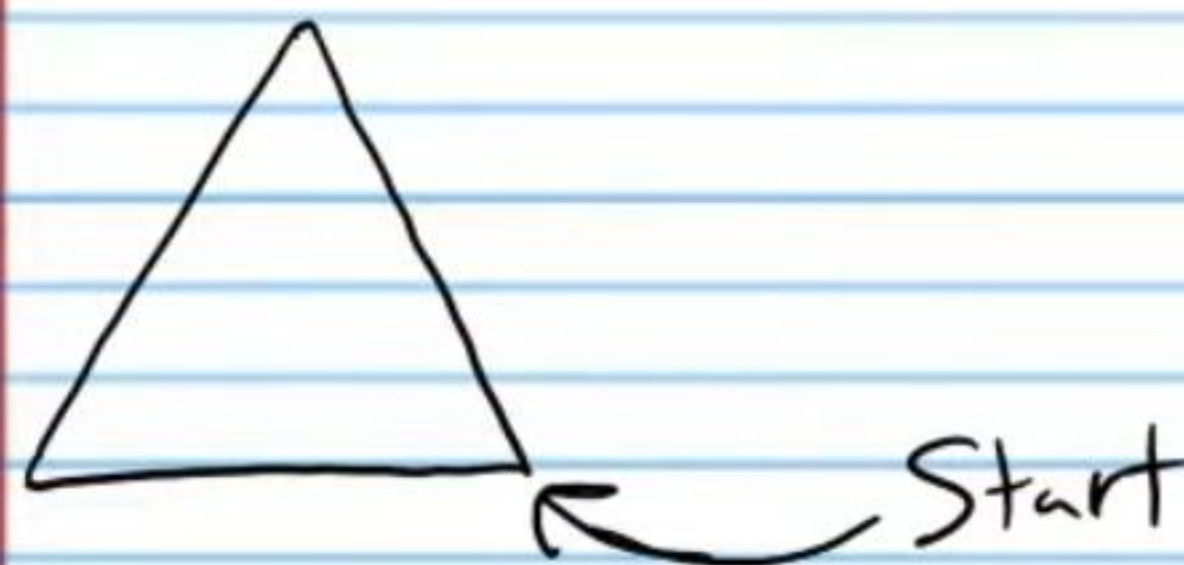
4B) An object travels on the sides of the shape below. It takes the object two minutes to move from one corner to the next. If the object starts on the top corner and moves counterclockwise, on which side will it be after moving for 195 minutes?



④ An object travels on the sides of the shape below. It takes the object five seconds to move from one corner to the next. If the object starts on the left corner and moves clockwise, on which side will it be after moving for 211 seconds?



④ An object travels on the sides of the shape below. It takes the object nine seconds to move from one corner to the next. If the object starts on the right corner and moves counterclockwise, on which side will it be after moving for 734 seconds?



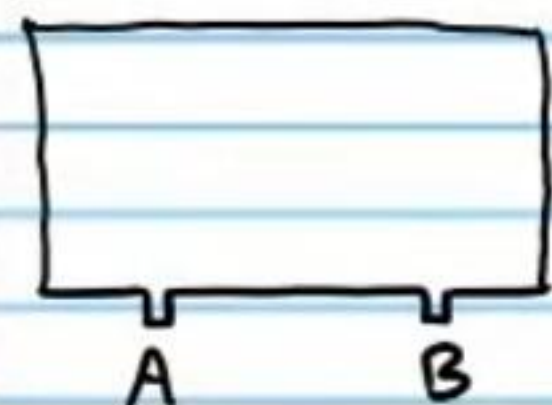
⑤A If Bill threads 3 square inches of clothing material every hour, and Lana threads 4 square inches every hour, how many square inches of clothing material will they thread in 12 hours if they work together?

⑤B A charity raises \$2,400 per month. A second charity raises \$4,000 per month. If the charities combine their funds, how much money will they raise in 8 months?

⑤C Katie and Donny are drinking from the same soda glass with two straws. If Katie drinks one ounce of soda per minute and Donny drinks three ounces of soda per minute, how long will it take them to drink 10 ounces of soda?

⑤D A toy company has a machine that makes stuffed bears. It makes 25 bears each hour. If the company buys an additional machine that makes 30 bears per hour, how long will it take the company to make 400 bears?

⑥A Below is an image of a water tank. If drain A is opened, the tank will drain completely in five minutes. If drain B is opened instead of drain A, the water will drain in six minutes. If both drains are opened instead of the first two scenarios, how long will it take for all the water to be drained?



⑥B There is a dry lakebed. There are two dammed up rivers that can fill the lake. One of the rivers can fill the lake in 5 days. The other river can fill the lake in 10 days. If both the dams are opened, how long will it take to fill the lake?

⑥C An auto mechanic wants to drain the coolant in a car. If he removes a drain plug, the coolant will drain completely in nine seconds. If he removes a hose instead, the coolant will drain completely in six seconds. How fast will the coolant drain if the mechanic removes the drain plug and the hose?

⑥D A homeowner can fill his backyard pool in seven hours if he uses his water hose, but if he uses his neighbor's water hose, the pool will be filled in eight hours. How long will it take for this homeowner to fill his pool if he uses both hoses?

⑦A A business is owned by five people, each with an equal share, and you are one of these people. If a shareholder sells you $\frac{1}{2}$ of his share and another shareholder keeps $\frac{3}{4}$ of his share and sells you the rest, what fraction of the business will you own?

78) A pizza is divided into eight slices. Each person in a group of four takes two pieces. If you are one of these people and one of the other individuals gives you $\frac{1}{6}$ of his share of the pizza, and another person gives you $\frac{1}{3}$ of one of his slices, what fraction of the entire pizza will you have?

79) 10 people are commissioned to paint a mural on the wall of a city monument. Each person will paint an equal area of the wall. If one person gets sick after finishing only $\frac{1}{4}$ of his work, and another person leaves for another job after finishing only $\frac{4}{5}$ of his work, what fraction of the entire wall will you have to paint if you are one of the 10 people and you have to finish your portion of the wall and also the unfinished portions of the two people who left?

80) A bottle contains two chemicals, chemical A and chemical B. $\frac{1}{4}$ of the bottle contains chemical A. After sitting on a shelf for three years, $\frac{1}{3}$ of chemical B converts to chemical C and $\frac{1}{2}$ of chemical A converts to chemical B. What fraction of the entire bottle is now chemical B?

81) A kitchen floor is 8.2 feet by 9.4 feet. If it costs \$5 to replace one square foot of the floor, how much will it cost to replace the entire kitchen floor?

82) Nine cubic meters of water are poured into a tank every minute. If the tank is 3.2 meters by 8 meters by 9 meters, how long will it take for the tank to be filled?

83) John needs to install concrete to make a driveway. He needs to fill an 85 feet by 20 feet rectangular area. If one bag of concrete covers 10 square feet of driveway, how many bags will John need to buy?

8D) Lesley wants to fill her fish tank with water. If the tank measures 5 feet by 2 feet by 2 feet, and water costs 9 cents per cubic foot, how much will it cost for her to fill the tank?

9A) Find 10% of 25%.

9B) What is 7.3% of 54%?

9C) Find 50% of 40%.

9D) What is 23% of 0.39%?

10A) Ricky drove 60 miles per hour for two hours. He then increased his speed to 65 miles per hour and drove for one hour at this new speed. What was his average speed for the entire trip?

10B) A horse runs 5 meters per second for 6 seconds, then slows to 4 meters per second and sustains this new speed for 4 seconds. What was the horse's average speed during his full run?

10C) Billy walked 3 miles per hour for 3 hours. He then jogged 10 miles per hour for 2 hours. What was his average speed during his movement?

10D) A space shuttle flies 1,200 mile per hour for 100 miles, then increases its speed to 1,500 miles per hour for 200 miles. What was its average speed for the entire travel?